EXAMPLE 13

Base-Isolated, Two-Story, 3D Frame - Nonlinear Time History Analysis

Problem Description

This is a two-story, three-dimensional frame with base isolation. The structure is subjected to earthquake motion in two perpendicular directions using the Loma Prieta acceleration records.

Hysteretic base isolators of the type described in Nagarajaiah et al. (1991) are modeled using the ETABS ISOLATOR1 elements, which show biaxial hysteretic characteristics.

Geometry, Properties and Loads

The structure is modeled as a single reinforced concrete frame with nine column lines and twelve bays. The floor slab is taken to be 8 inches thick, covering all of the specified floor bays at the base and the 1st story level. At the second story level the corner column as well as the two edge beams are eliminated, together with the floor slab, to render this particular level unsymmetric, as depicted in Figure 13-1.

A modulus of elasticity of 3000 ksi is used. The self-weight of concrete is taken as 150 pcf. Kip-inch-second units are used.

The geometry of the structure is shown in Figure 13-1.

Technical Features of ETABS Tested

- Three-dimensional frame analysis
- Use of area (floor) objects
- Use of biaxial hysteretic elements
- Point assignments
- Nonlinear time history analysis using ritz vectors

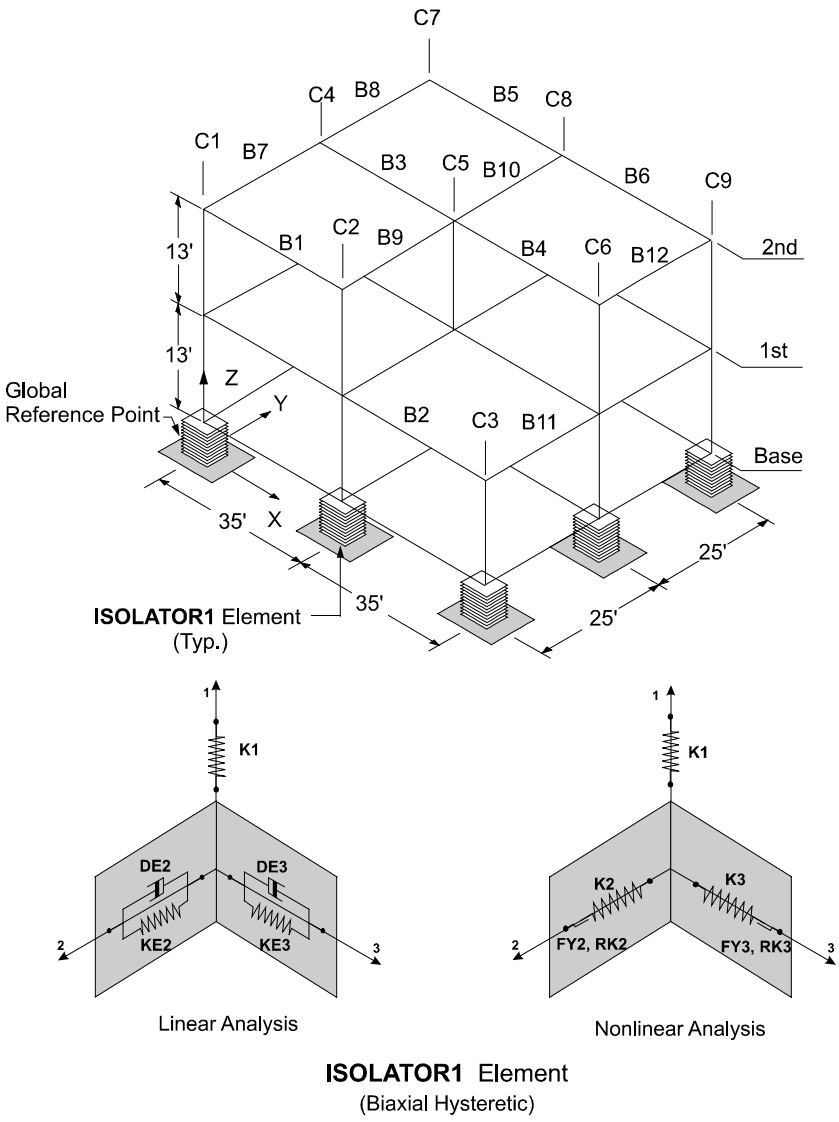


Figure 13-1 Base-Isolated Three-Dimensional Frame

Results Comparison

The example frame analyzed using ETABS is also analyzed using SAP2000 (Computers and Structures 2002) for time history loads (SAP2000 has been verified independently). A comparison of key ETABS results with SAP200 is presented in Table 13-1.

Table 13-1 Comparison of Results for Time History Analysis

Quantity	ETABS	SAP2000
Maximum Uy Displacement, Column C9 at 2nd Floor	3.4735	3.4736
Maximum Axial Force, Column C1 at base	13.56	13.55

A typical output produced by the program is shown in Figure 13-2. It shows the load-deformation relationship in the major direction for a typical isolator member.

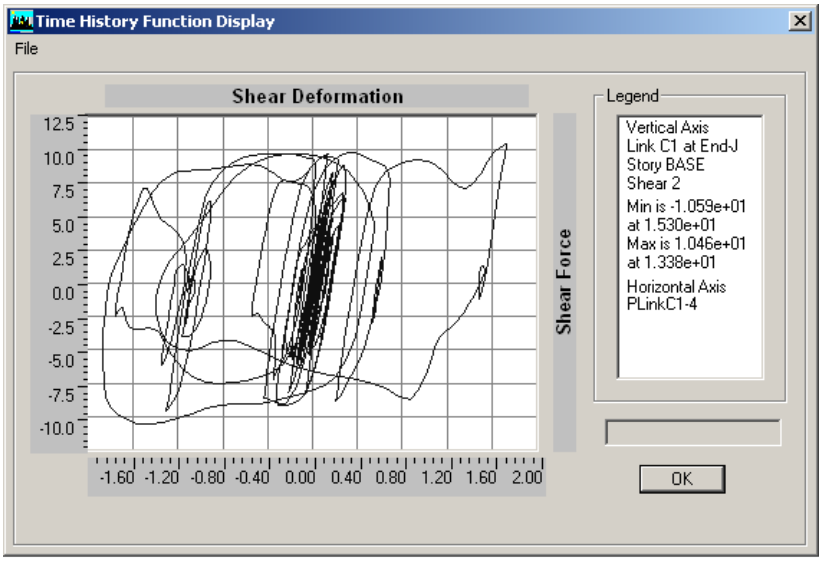


Figure 13-2 Load Deformation Diagram

Computer Files

The input data file for this example is Example 13.EDB. The time history files are LP-TH0 and LP-TH90. All of these files are provided as part of the ETABS installation.

Conclusion

The results comparison shows essentially exact agreement between ETABS and SAP2000.